High Level Problems (HLP)

SUBJECTIVE QUESTIONS

1. A current of 4 A flows in a coil when connected to a 12 V d.c. source. If the same coil is connected to a 12 V, 50 rad/s, AC source, a current of 2.4 A flows in the circuit. Determine the inductance of the coil. Also, find the power developed in the circuit if a 2500 μF condenser is connected in series with coil.
[REE - 1993]

2. A box P and a coil Q are connected in series with an AC source of variable frequency. The EMF of source is constant at 10 V. Box P contains a capacitance of 1 μF in series with a resistance of 32 Ω. Coil Q has a self inductance 4.9 mH and a resistance of 68 Ω. The frequency is adjusted so that the maximum current flows in P and Q. Find the impedance of P and Q at this frequency. Also find the voltage across P and Q respectively. **[REE - 1998]**

3. In a series LCR circuit with an ac source of 50 V, R = 300Ω, frequency $\nu = \frac{50}{\pi}$ Hz. The average electric field energy, stored in the capacitor and average magnetic energy stored in the coil are 25 mJ and 5 mJ respectively. The RMS current in the circuit is 0.10 A. Then find:
(a) Capacitance (c) of capacitor
(b) Inductance (L) of inductor.
(c) The sum of rms potential difference across the three elements.

4. An inductor 20×10^{-3} Henry, a capacitor 100 μF and a resistor 50Ω are connected in series across a source of EMF V = 10 sin 314 t. Find the energy dissipated in the circuit in 20 minutes. If resistance is removed from the circuit and the value of inductance is doubled, then find the variation of current with time (t in second) in the new circuit. **[REE - 1999]**

5. The electric current in an AC circuit is given by $i = i_0 \sin \omega t$. What is the time taken by the current to change from its maximum value to the rms value? **[REE - 1999]**

6. A circuit containing a 0.1 H inductor and a 500 μF capacitor in series is connected to a 230 volt, 100/π Hz supply. The resistance of the circuit is negligible. (a) Obtain the current amplitude and rms values. (b) Obtain the rms values of potential drops across each element. (c) What is the average power transferred to the inductor? (d) What is the average power transferred to the capacitor? (e) What is the total average power absorbed by the circuit? ['Average' implies average over one cycle.]

7. A series LCR circuit with L = 0.125/π H, C = 500/π nF, R = 23 Ω is connected to a 230 V variable frequency supply.
(a) What is the source frequency for which current amplitude is maximum? Obtain this maximum value.
(b) What is the source frequency for which average power absorbed by the circuit is maximum? Obtain the value of this maximum power.
(c) For what reactance of the circuit, the power transferred to the circuit is half the power at resonance? What is the current amplitude at this reactance ?
(d) If ω is the angular frequency at which the power consumed in the circuit is half the power at resonance, write an expression for ω
(e) What is the Q-factor (Quality factor) of the given circuit?

8. The maximum values of the alternating voltages and current are 400 V and 20 A respectively in a circuit connected to 50 Hz supply and these quantities are sinusoidal. The instantaneous values of the voltage and current are $200\sqrt{2}$ V and 10 A respectively. At that instant both are increasing positively. Determine the average power consumed in the circuit.

9. A 750 Hz, 20 V source is connected to a resistance of 100 Ω, a capacitance of 1.0 μF and an inductance of 0.18 H in series. Calculate the following quantities : **[Olympaid 2013-14]**

(a) Impedence of the circuit

(b) Draw an impedence diagram with suitable scale

(c) Power factor

(d) The time in which the resistance will get heated by 10ºC, provided that the thermal capacity of resistance = 2 J/ºC

10. In the given circuit

Calculate

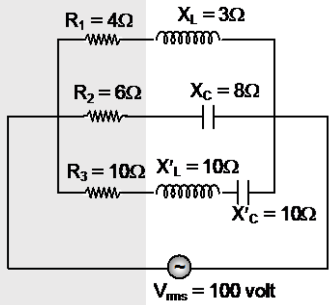
(a) Current in each branch.

(b) Power generated in each resistance.

(c) Total power generated in the circuit.

(d) Net current drawn from source.

(e) Net impedance of the circuit.



11. A metallic coil of N turns of radius a, resistance R, and inductance L is held fixed with its axis along a spatial uniform magnetic field $\vec{B}$ whose magnitude is given by $B_0 \sin(\omega t)$.

(a) Write the emf equation for the current i in the coil.

(b) Assuming that in the steady state i. oscillates with the same frequency ω as the magnetic field, obtain the expression for i.

(c) Obtain the force per unit length. Further obtain its oscillatory part and the time-averaged compressional part.

(d) Calculate the time-averaged compressional force per unit length given that B_0 = 1.00 tesla, N = 10, a = 10.0 cm, ω = 1000.0 rad-s^{-1}, R = 10.0 Ω , L = 100.0 mH.

HLP Answers

1. .08 H; 17.28 watt **2.** P = 76.96 Ω, Q = 97.59Ω, P ≈ 7.7 V; Q = 9.8 V, net impedance =100 Ω

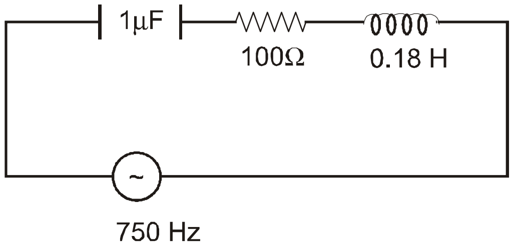
3. (a) C = 20 μF, (b) 1 H, (c) 90 V **4.** 951.52 J; 0.52 cos 314 t **5.** T/8 or $\frac{\pi}{4\omega}$

6. (a) $23\sqrt{2}$ A, 23 A, (b) 460 volt, 230 volt, (c) zero, (d) zero, (e) zero

7. (a) 2000 Hz, $10\sqrt{2}$ A, (b) 2000 Hz, 2300 watt, (c) 23 Ω, 10 A. (d) $\frac{0.125}{\pi}\omega - \frac{1\times10^9}{\omega\frac{500}{\pi}} = \pm\ 23$ (e) 500/23

8. P = 3864 W

9. $z = \sqrt{(X_L - X_C)^2 + R^2}$



$\omega = 2n\pi = 2(750)\pi = 1500\pi$

$X_L = \omega_L = (1500\pi) \quad (0.18) = 848.5 \quad \Omega$

$X_C = \frac{1}{\omega C} = \frac{10^6}{1500\pi} = 212.12 \quad \Omega$

$z = \sqrt{(636.4)^2 + (100)^2} = 100\sqrt{(6.364)^2 + 12}$

$= 100 \times 6.44 \cong 644 \quad \Omega$

$\tan\phi = \frac{X_L - X_C}{R} = \frac{848.5 - 212.12}{100} = 6.36$

(c) $\cos\phi = \frac{R}{z} = \frac{100}{644} = 0.155$

(b) impedence is constant as n is constant

(d) $i_{rms} = \frac{\varepsilon_{rms}}{z} = \frac{20}{644}$ A

$H = (i_{rms})^2 Rt = \left(\frac{20}{644}\right)^2 (100)t \quad \Rightarrow (ms)\ (\Delta\theta) = \left(\frac{20}{644}\right)^2 (100)\ t$

$(2)\ (10) = \left(\frac{20}{644}\right)^2 (100)\ t \quad \Rightarrow t = \left(\frac{644}{20}\right)^2 \times \frac{1}{100} \times 20 \Rightarrow t = 207.36$ sec.

10. Impedence of each branch $Z_1 = \sqrt{R_1^2 + X_L^2} = \sqrt{4^2 + 3^2} = 5$

$Z_2 = \sqrt{6^2 + 8^2} = 10$

$Z_3 = \sqrt{(10)^2 + (10 - 10)^2} = 10$

(a) Current in each branch $I_1 = \frac{V}{Z_1} = \frac{100}{5} = 20$ Amp.

$I_2 = \frac{100}{10} = 10$ Amp.

$I_3 = \frac{100}{10} = 10$ Amp.

(b) Power in each branch.

$P_1 = (I_1)^2 R_1 = (20)^2 (4) = 1600$ watt

$P_2 = (I_2)^2 R_2 = (10)^2 (6) = 600$ watt

$P_3 = (I_3)^2 R_3 = (10)^2 (10) = 1000$ watt

(c) Net power of the circuit.

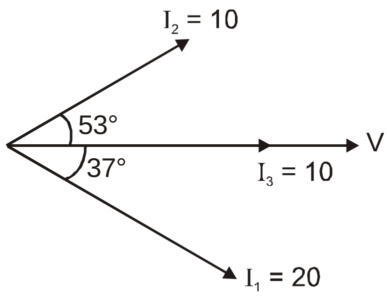
$P = P_1 + P_2 + P_3 = 3200$ watt

phase difference between voltage & current in each branch.

$\tan\phi_1 = \frac{X_L}{R_1} = \frac{3}{4} \Rightarrow \phi_1 = 37°$

$\tan\phi_2 = \frac{X_C}{R_2} = \frac{8}{6} \Rightarrow \phi_2 = 53°$

$\tan\phi_3 = 0 \Rightarrow \phi_3 = 0$



(d) Net current drawn from source.

$\sqrt{(I_1\cos 37° + I_2\cos 53° + I_3)^2 + (I_1\sin 37° - I_2\sin 53°)^2} = \sqrt{1040}$

(e) Net impedance of the circuit. $Z = \frac{V}{I} = \frac{100}{\sqrt{1040}}$

11. (a) $iR + L\frac{di}{dt} = -N\pi a^2 B_0 \omega \cos \omega t$

(b) $i = \frac{N\pi a^2 B_0\omega \left(R\cos\omega t + \omega L \sin\omega t\right)}{R^2 + \omega^2 L^2}$

(c) $\frac{dF}{d\ell} = -\frac{NB_0^2 \pi a^2 \omega}{R^2 + \omega^2 L^2}$ $(R \sin \omega t \cos \omega t + \omega L \sin^2 \omega t)$

$= \left.\frac{dF}{d\ell}\right|_{av} = -\frac{NB_0^2 \pi a^2 \omega^2 L}{2(R^2 + \omega^2 L^2)}$

$= \left.\frac{dF}{d\ell}\right|_{osc} = -\frac{NB_0^2 \pi a^2 \omega}{2(R^2 + \omega^2 L^2)}$ $(R \sin 2\omega t - \omega L \cos 2\omega t)$

(d) $\left.\frac{dF}{d\ell}\right|_{av} = 1.55\ N.m^{-1}$